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Literature Review

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Article

Hidden states and dynamics of fractional fillings in twisted $MoTe_2$ bilayers

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Transient pump-probe + RMCD find the hidden correlated FQAH states in twisted bilayer MoTe2;

Measure and analyze the dynamic process in electron doping and hole doping state.

Nature 641, 1149–1155 (2025)

OUTLINE

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1. Background

2. Measurement

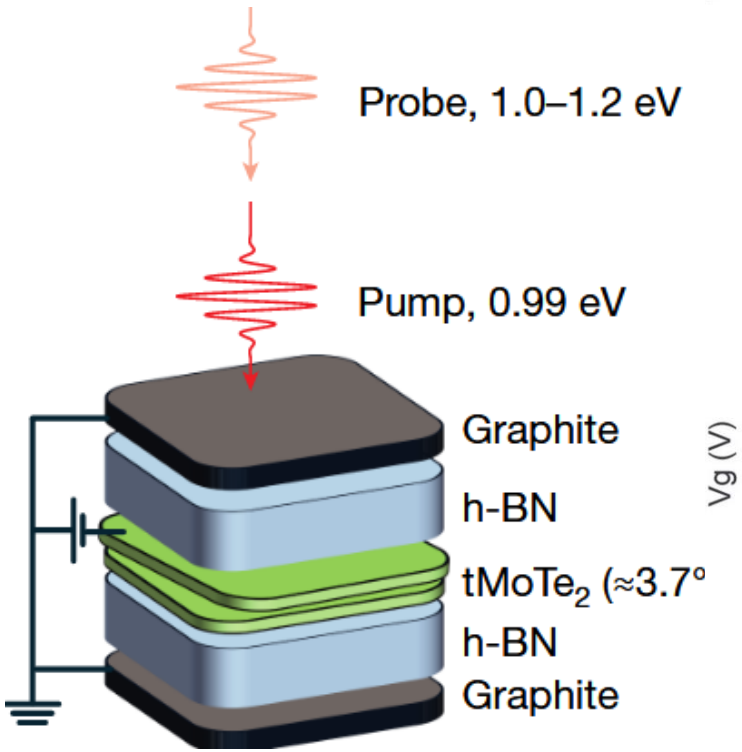
3. Summary

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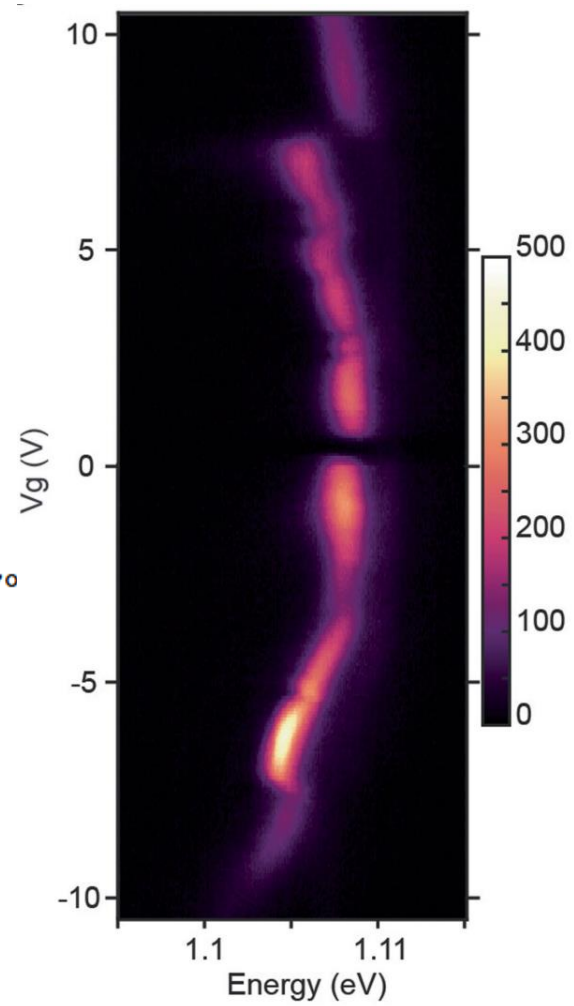
01.

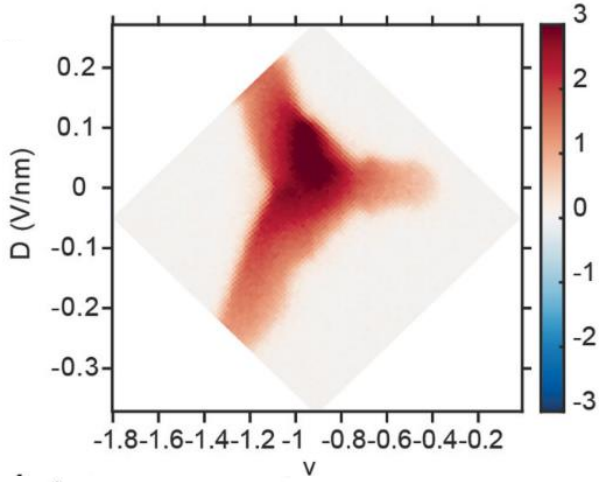
Background

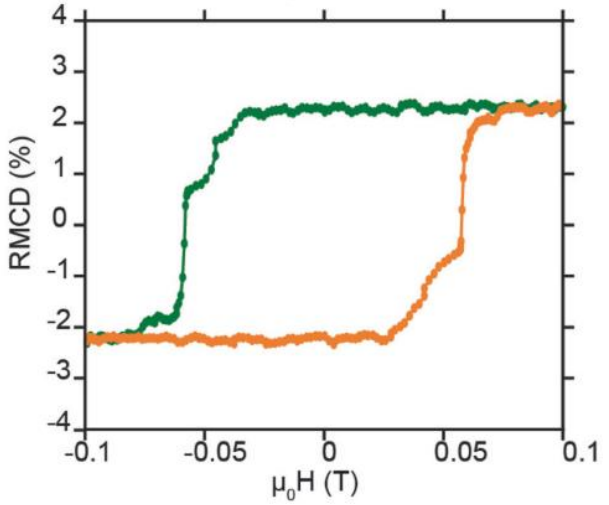
Device characterization



Dual-gated R-stacked MoTe2 bilayers

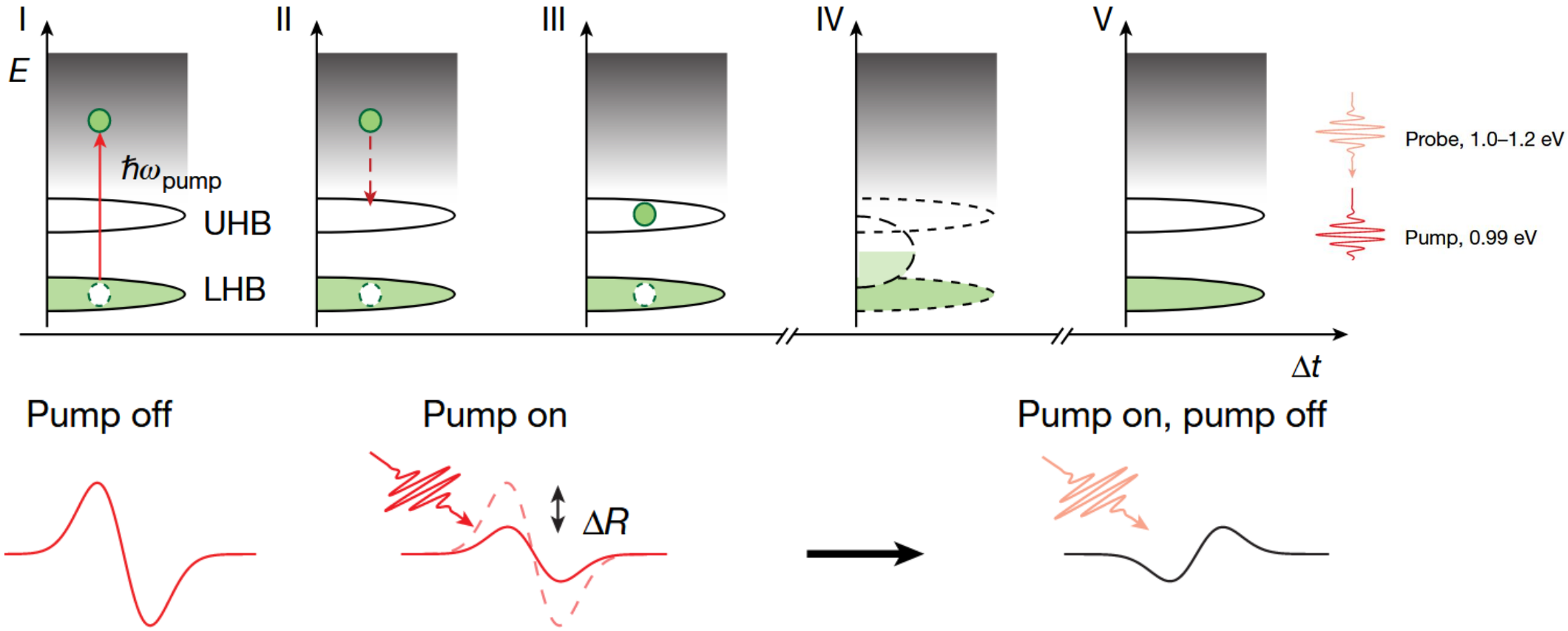






Transient pump-probe

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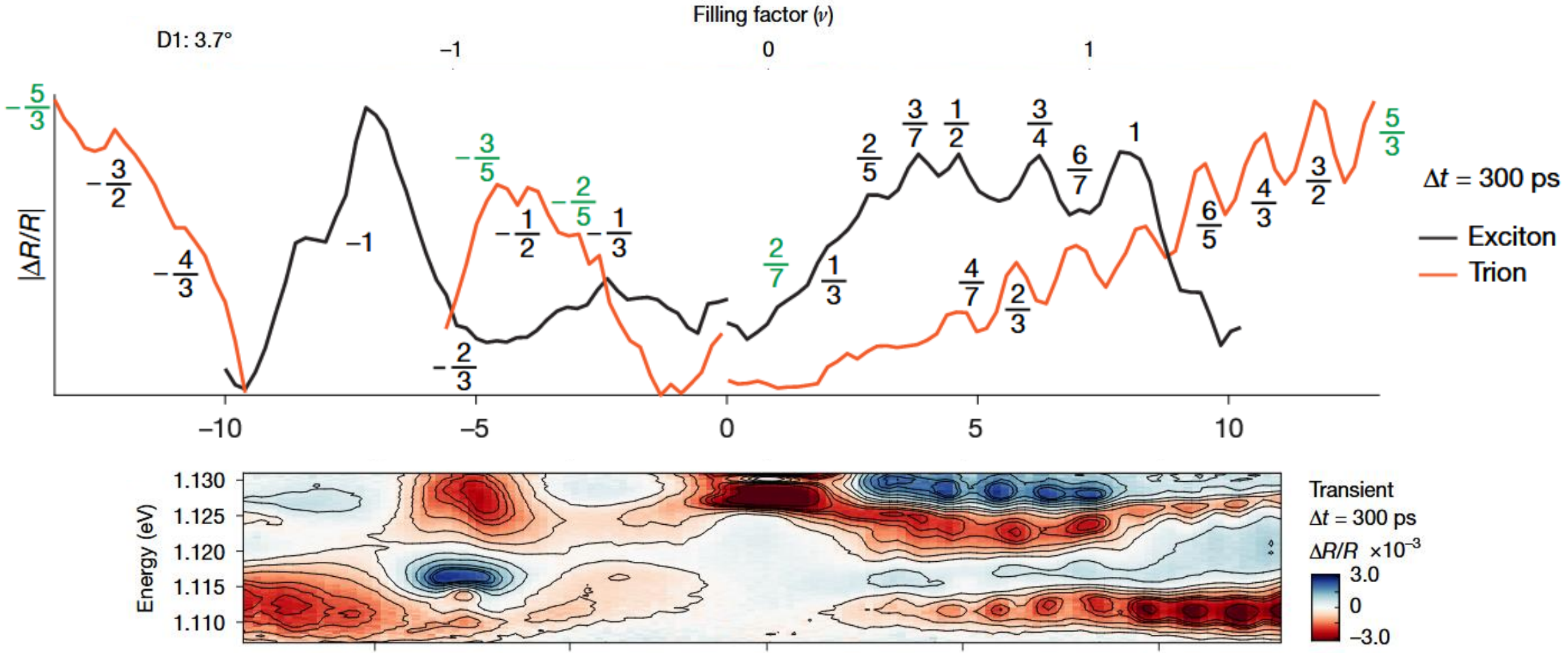
(ΔR/R) Detect the change in effective dielectric constant and the resulting decrease in oscillation strengths

02.

Measurement

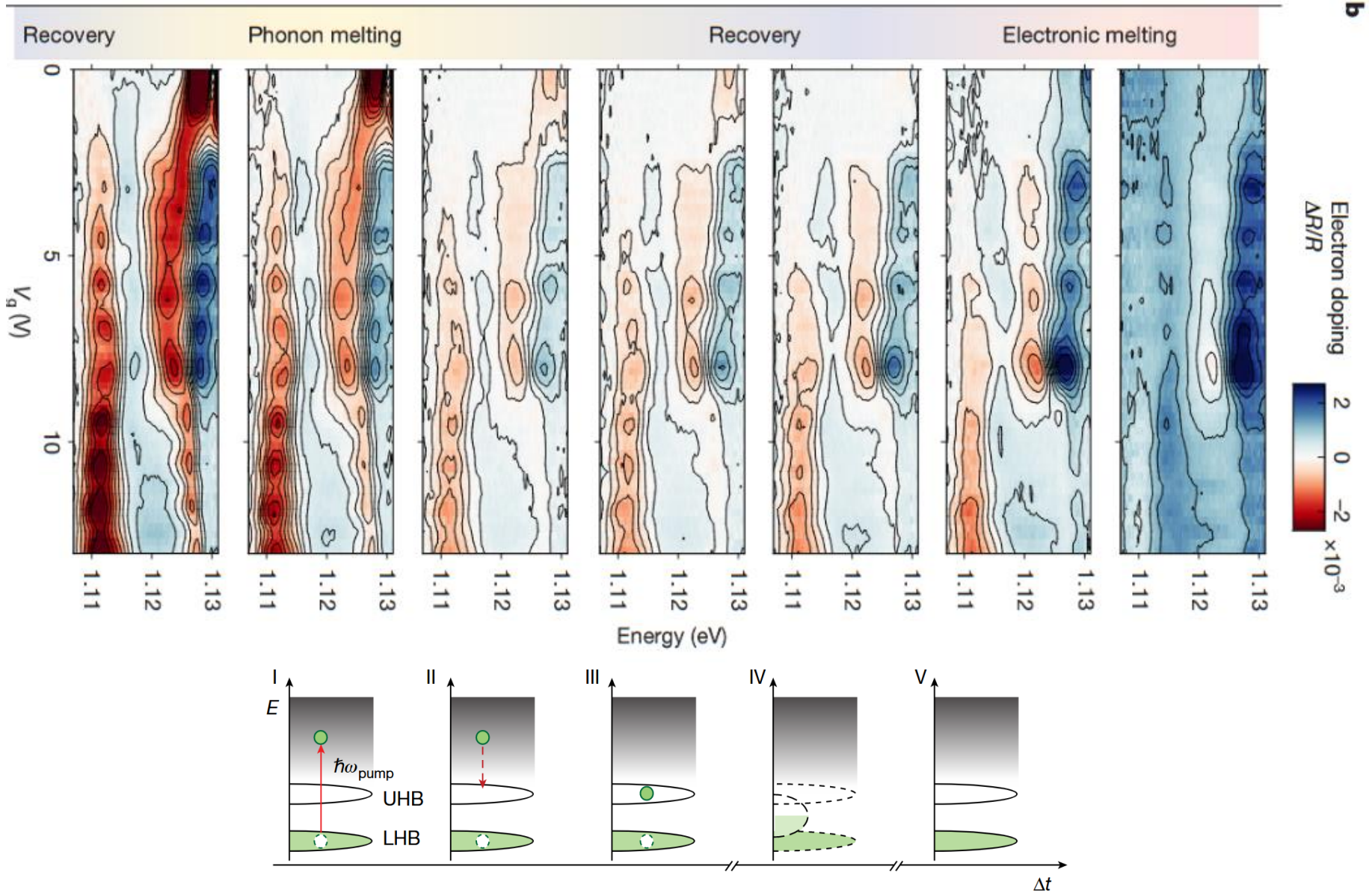
RMCD measurement

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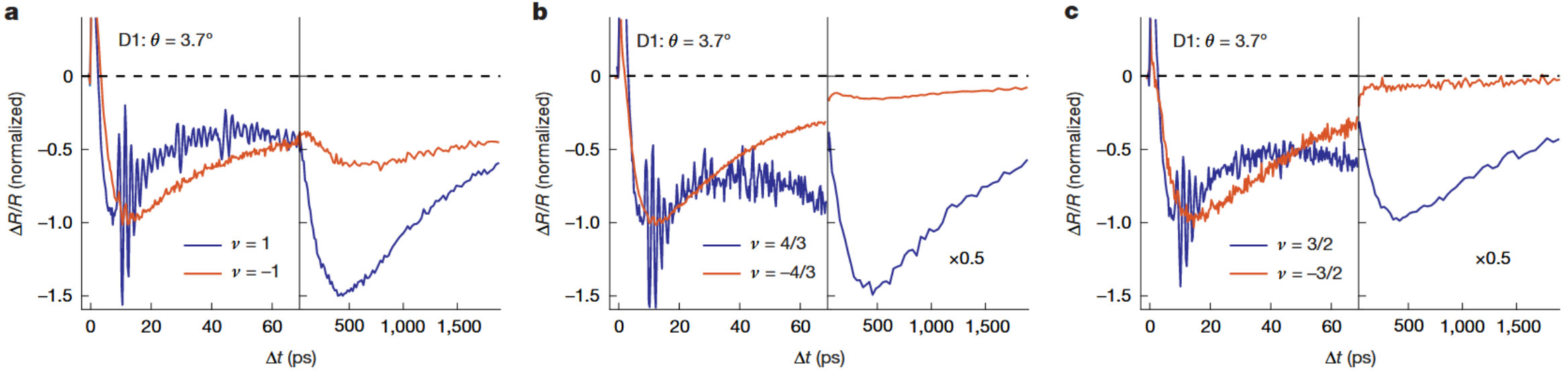
Exciton (about 1.120–1.135 eV) and trion (about 1.100–1.120 eV)

Dynamic process



Dynamic process explanation

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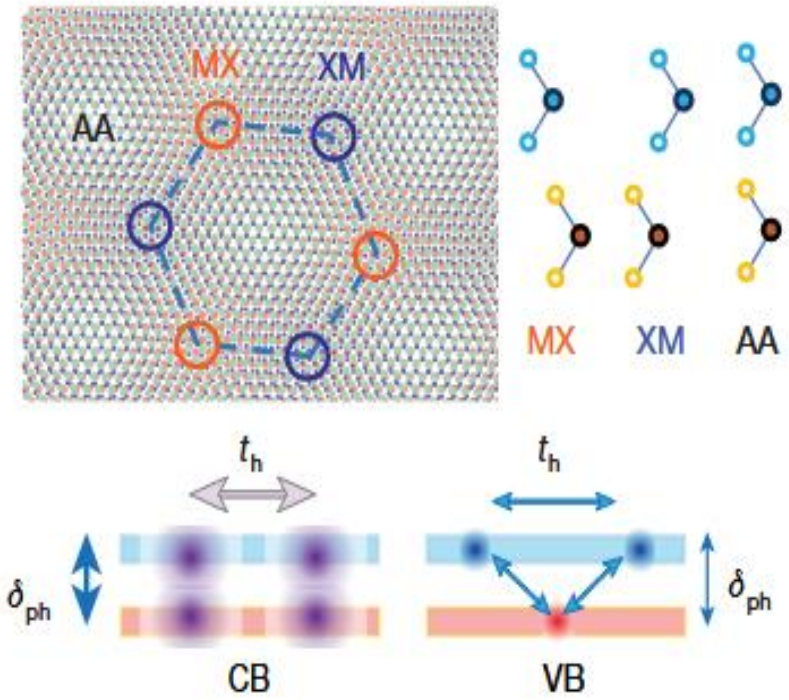


Electron doping initially decays 2.0–3.5 ps and recoveries with 20–40 ps.
This stage is due to disordering from charge hopping, called ‘electronic melting’.

The second minima is time-delayed by 150 ps, reflecting the time for reheating the electrons by phonons. This remelting of states is called ‘phonon melting’.

Interlayer tunnelling in moiré

$$H_\perp = t_\perp c^\dagger_{\text{top}} c_{\text{bottom}} + \text{h. c.}$$



The key feature of twisted transition-metal dichalcogenide bilayers is the layer skyrmion texture of interlayer tunnelling

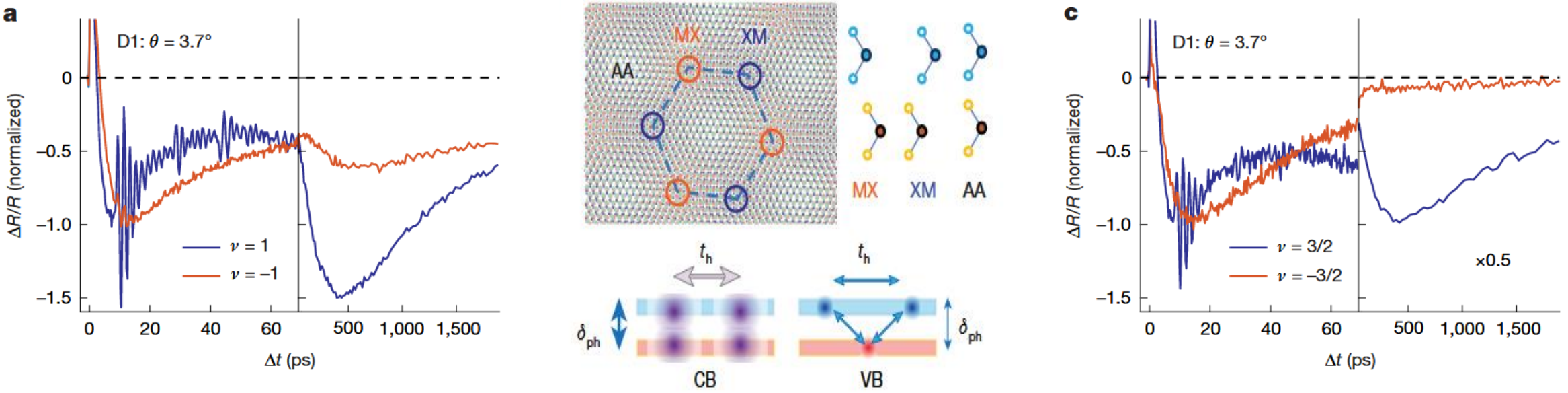
Interlayer tunnelling is large in the AA regions but vanishes in the MX/XM regions.

For electron doping, the moiré conduction band derive from the Q point owing to interlayer hybridization and are delocalized spatially in zigzag stripes across moiré unit cell.

For hole-doped states, the uppermost moiré valence band near the K points is localized to the MX or XM regions that form the honeycomb lattice.

“Melting” process explanation

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In conduction band (electron-doped), the breathing phonon modulate δ_{ph} which affects vertical hopping (interlayer tunnelling amplitude) explaining the “phonon melting” exists in electron doped instead of hole doped side (for MX, XM there is less overlapping).

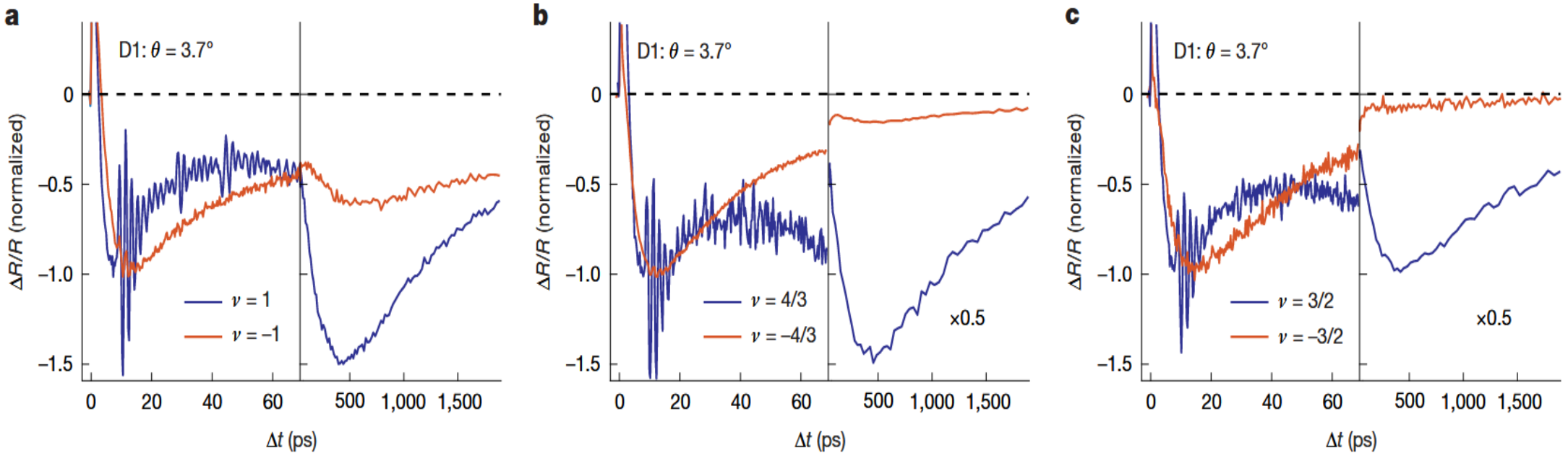
In valance band (hole-doped), the Holon/doublon hopping is inhibited between neighbouring MX/XM sites and is also small between next-nearest-neighbour sites, thus slowing “electron melting” and recovery dynamics.

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03.

Summary

Summary



In summary, in this work they use transient pump probe to find and analyze dynamic process of the hidden correlated FQAH states in twisted bilayer MoTe2;

They get conclusion that the process is related to the oscillation strength of Interlayer tunnelling in different spacial moiré structure with conduction band and valance band.



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THANKS

Speaker: Zhongqi Xiu